Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/11/2011
PI Name:	Wise, John Pierce Ph.D.		
Project Title:	Mechanisms of Altered Gravity Potentiation of Metal-Induced Genotoxicity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	John.Wise@usm.maine.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	207-228-8050
Organization Name:	University of Southern Maine		
PI Address 1:	96 Falmouth St		
PI Address 2:	PO Box 9300		
PI Web Page:			
City:	Portland	**State:**	ME
Zip Code:	04103-4864	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	07/01/2009	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received extension to 6/30/2012 per A. Chu/ARC (Ed., 4/29/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AL11G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The long-term objective of this research is to develop a model of the effects of microgravity on the cellular response to genotoxicity. The effects of microgravity on genotoxic response is important because NASA astronauts and personnel in microgravity will be exposed to significant genotoxic agents including chemicals (e.g. metals) and radiation and the potential for microgravity to exacerbate the genotoxic outcome is poorly understood. The short term objective of this research is to initially focus on chromium (Cr) because it is one of the only metals with a well-established DNA repair response to genotoxicity that is also a component of lunar soil. We will investigate the central hypothesis that altered gravity potentiates metal ion- and particle-induced genotoxicity. We will test this hypothesis through two Specific Aims: 1) Determine Altered Gravity Effects on Metal-induced DNA Damage Formation and Repair in Human Cells and 2) Determine if Altered Gravity Effects Are Conserved Across Cell Types and Species. These data when combined with our data generated by our studies of lunar dust in cells at normal gravity can be used to help determine appropriate exposure limits to moon dust. Our data will also be the first to evaluate how altered gravity affects metal-induced DNA repair gene expression and signal transduction as well as effects on metal-induced apoptosis. These data will also indicate which responses are general and which are cell or species specific. Altogether, these data should provide important insight into the mechanisms of altered gravity on metal-induced genotoxicity and suggest possible approaches for potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We found that cells grown on microcarriers and rotated at 23 rpm had less damage than cells grown at normal gravity under normal culture conditions. With the wide range of speed settings on the HARV culture vessel rotating base it is possible rotational speed may play a role in the chromosome damage results at 23 rpm in the previous experiments. To test for this we conducted experiments at 8 rpm results suggest that lower rotational speeds have a larger effect. In order to further explain the possible differences seen in the amount of damage seen in cells treated in microgravity versus normal gravity, we conducted chemical uptake experiments to determine if the difference in damage was due to a difference in chromium uptake under differing gravity conditions. We found that cells in microgravity take up less chromium than cells under normal gravity conditions in human lung cells. When the chromosome damage results are corrected for chemical uptake we see that the amount of chromium-induced chromosome damage is similar. These data were consistent when tested under the different rotational speeds (8 rpm vs 23 rpm). This year we also conducted experiments in human skin cells and found similar results as those found in human lung cells. We found that there is less damage in skin cells exposed to microgravity when also exposed to sodium chromate than in cells grown at normal gravity. Additionally, we conducted uptake experiments in the human skin cells. We found that like the human lung cells there is less intracellular chromium present in the cells under microgravity conditions. When the level of chromosome damage seen is compared based on intracellular chromium concentration, the amount of damage under both microgravity and normal gravity conditions is the same. In addition to working on the effects of microgravity we extended our investigation this year to consider the contribution of hypergravity. We considered the effect of hypergravity on the cytotoxic effect of sodium chromate, a soluble form of chromium, in human lung cells. Hypergravity exacerbated the effect of sodium chromate-induced cytotoxicity at lower concentrations in lung cells, however, at the highest concentration there was no difference in effect. We also looked at these effects in human skin cells. In skin cells there was no difference in the effect of hypergravity on sodium chromate-induced cytotoxicity. We also considered the contribution of hypergravity on the genotoxic effects of sodium chromate treatment. These experiments are currently under analysis. We began studies considering another hexavalent chromium compound, lead chromate. Lead chromate is an insoluble form of chromate and is administered as a particle in suspension in cell cultures. We were unsure of how these studies would work as the particles come into direct and continuous contact with cells in the normal gravity conditions but not in the microgravity conditions. To address this we also treated cells attached to microcarriers under normal gravity conditions. The data suggest that there is no difference in the cytotoxic effect of lead chromate between the microgravity and normal gravity conditions, however, we did have high variability between the 2 experiments, likely due to the added component of the particle. Additional experiments will be done to address this issue and further resolve the data. We have begun to investigate these effects in other species, specifically in bowhead whale and medaka cells. For these cells, we focused on the effect of hypergravity. We found that the cytotoxic effect of hexavalent chromium is not exacerbated by hypergravity in bowhead whale lung cells at 10g, however when we tested these cells at 2g we found a much greater difference. We also considered the effects of chromate in medaka cells. We found that, similar to human skin cells, hypergravity did not exacerbate the chromium-induced cytotoxic effect in medaka cells. We then considered the genotoxic effect of chromium in medaka cells exposed to hypergravity. There was less chromium-induced damage seen in the cells exposed to hypergravity than cells exposed to normal gravity.
Bibliography Type:	Description: (Last Updated: 08/24/2017)
Abstracts for Journals and Proceedings	Wise JP Jr, Wheeler C, Wise JP Sr. "The Impact of Altered Gravity on Chromate Genotoxicity: Using a Bioreactor to Simulate Microgravity in Cell Culture." 2011 Annual Meeting and ToxExp (TM) of the Society for Toxicology, Washington, DC, March 6-10, 2011. Toxicologist (supplement to Toxicological Sciences), 2011 Mar;120(Suppl 2):60. http://www.toxicology.org/AI/Pub/Tox/2011Tox.pdf , Mar-2011
Abstracts for Journals and Proceedings	Braun M, Wheeler C, Shehata T, Wise JP Sr. "Using Microcarriers and a Bioreactor to Support Cell Growth in a Simulated Microgravity Environment." University of Southern Maine's 8th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April 29, 2011. University of Southern Maine's 8th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April 29, 2011. , Apr-2011

Abstracts for Journals and Proceedings	Shively E, Wise JP Jr, McKay J, Wise JP Sr. "The Effect of Hypergravity on Cr(VI)-induced Cytotoxicity and Genotoxicity." University of Southern Maine's 8th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April 29, 2011. University of Southern Maine's 8th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April 29, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Wise JP Jr, Wheeler C, Wise JP Sr. "The Impact of Altered Gravity on Chromate Genotoxicity: Using a Bioreactor to Simulate Microgravity in Cell Culture." University of Southern Maine's 8th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April 29, 2011. University of Southern Maine's 8th Annual Thinking Matters Conference: USM Research, Scholarship, and Creativity Symposium, April 29, 2011. , Apr-2011